



Innovative Interposer Solutions for HBM3/4: A Path to 12.8Gbps

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Abstract

The AI boom is driving the memory industry towards HBM4 at 12.8Gbps, doubling Interface Width from 1024 to 2048. Semiconductor foundries are enhancing 2.5D and 3D packaging solutions for seamless HBM3 & HBM4 to SoC integration, addressing layout and SI-PI challenges due to increased crosstalk and loss from higher I/O counts and deeper micro bump pins. Achieving 12.8Gbps HBM3/4-to-SoC integration requires systematic optimization of layout and SI-PI aspects.

This paper demonstrates an efficient SI-PI layout optimization methodology, developed alongside a novel crosstalk shielding structure [1][10] that significantly reduces crosstalk, enables higher speeds for HBM3 and HBM4. Our methodology is novel because it:

1. Leverages 2D and 3D-EM tools to balance accuracy and extraction time.
2. Characterizes the interposer with Voltage Transfer Function (VTF) based Insertion-Loss (IL), Power-Sum-Crosstalk (PSXT), and Insertion-Loss-to-Crosstalk Ratio (ICR).
3. Decomposes interposer-induced jitter into ISI, crosstalk, and rise-fall time degradation to identify the dominant channel parameter affecting EYE closure, facilitating better layout and I/O architecture optimization.

This paper addresses critical industry questions for achieving 12.8Gbps by evaluating Silicon (Si) & Organic (RDL) Interposers, determining optimal variants, stack-up layers, and routing pattern for HBM3/4-12.8Gbps. We provide a system-level jitter budget indicating available margin at 12.8Gbps and assess the need for I/O architecture enhancements such as equalization.

Author(s) Biography

Ilamparidhi I is a Principal SI-PI Engineer at Alphawave Semi with over 19 years of industry expertise in Analog Mixed Signal (AMS) System Modelling & Optimization, Advanced Package Design & Optimization, and PCB design. He leads a technical team focused on developing next-generation solutions for UCIE and HBM interfaces. Ilamparidhi holds an MTech in Microwave and Photonics from the Indian Institute of Technology, Kanpur (IIT-K), and has worked with leading companies such as Cisco, Seagate, Rambus, Xilinx, and Freescale Semiconductor. In the past year, he has driven key innovations at Alphawave Semi, resulting in the filing of four US patents (pending approval) related to advanced packaging design and Analog Mixed Signal (AMS) system for high-performance HBM and UCIE technologies.

Suriya Prakash T is a Senior SI-PI Engineer at Alphawave Semi with 8+ years of expertise in IP, advanced packaging, and PCB design. He specializes in Analog Mixed Signal (AMS) system modelling, simulations, and innovations in advanced packaging design for HBM and UCIE, with one US patent (pending approval).

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1. Introduction

In recent years, advancements in AI, ML, big data, and cloud computing have driven unprecedented demand for high-performance computing (HPC), requiring massive memory and bandwidth. Traditional DDR and GDDR memory cannot meet these demands due to bandwidth and capacity limitations. Consequently, 3D stacked high-bandwidth memory (HBM) based on TSV technology, with 1024 or more I/O ports, has emerged to bridge the performance gap between memory and processors. Industries focused on advanced packaging solutions using 2.5D and 3D technologies are striving to achieve 12.8Gbps lane speeds for HBM3/4 to SoC integration.

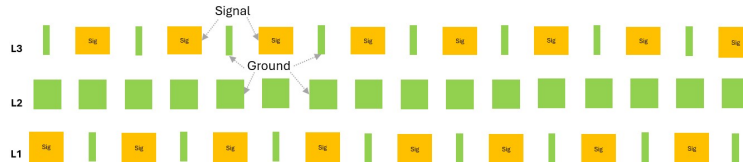


Fig1: Conventional Si Interposer signal ground routing pattern of HBM3 interface

Fig. 1 illustrates the cross-sectional view of a typical Silicon (Si) Interposer Ground-Signal-Ground (GSG) routing pattern [5][8] for HBM3 interfaces at speeds up to 7.2Gbps. It features two signal routing layers (L1 & L3) and one ground layer (L2), which serves as the return path for signals in L1 and L3 and provides crosstalk shielding. Ground shielding traces are placed between signal traces in L1 and L3 to further reduce crosstalk. The remaining metal layers are used for the interposer power distribution network (PDN).

To support higher data rates in HBM3/4, this paper explores achieving 12.8Gbps using Silicon (Si) and Organic (RDL) Interposer technologies. The conventional design (Fig. 1) was analyzed for higher speeds, focusing on VTF based insertion loss (IL), power-sum-crosstalk (PSXT), and Insertion-Loss-to-Crosstalk Ratio (ICR). System-level EYE simulations using IBIS models for the SoC-PHY and HBM-Memory in ADS Memory Designer (Fig. 3) were conducted to assess system margin at 12.8Gbps. This paper demonstrates an efficient SI-PI layout optimization methodology, developed alongside a novel crosstalk shielding structure [1][10] that significantly reduces crosstalk, enables higher speeds for HBM3 and HBM4. Our methodology is novel because it:

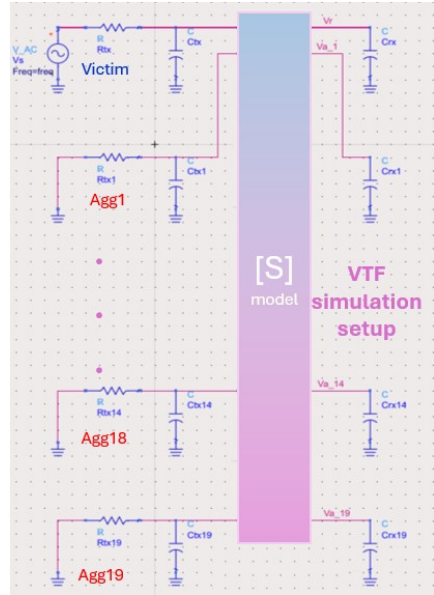
- Leveraging 2D and 3D-EM tools for a balance of accuracy and extraction time.
- Characterizing the interposer with VTF based Insertion-Loss (IL), Power-Sum-Crosstalk (PSXT), and ICR. VTF based metrics incorporate both resistive and capacitive components of Tx and Rx terminations
- Decomposing interposer-induced jitter into ISI, crosstalk, and rise-fall time degradation to identify the dominant channel parameter affecting EYE closure, facilitating better layout and I/O architecture optimization.

This paper addresses critical industry questions for achieving 12.8Gbps by evaluating Interposer variants (Si Vs. Organic), determining the optimal variant, stack-up layers, and

routing pattern for HBM3/4-12.8Gbps. We provide a system-level jitter budget indicating available margin at 12.8Gbps and assess the need for I/O architecture enhancements such as equalization.

1.1 Channel Characterization of Conventional Routing

The interposer channel can be characterized using three key parameters: VTF based insertion loss (VTF-IL), Power Sum Crosstalk (VTF-PSXT), and Insertion Loss to Crosstalk Ratio (VTF-ICR). By studying these parameters, we can predict the channel's performance, especially since VTF-ICR has a direct correlation with EYE opening.



$$VTF_{IL} = 20 \log_{10} \left| \frac{V_r(f)}{V_s(f)} \right|$$

$$VTF_{Xtalk} = 10 \log_{10} \left[\sum_{i=1}^{19} \left| \frac{V_{ai}(f)}{V_s(f)} \right|^2 \right]$$

$$VTF_{ICR} = VTF_{IL} - VTF_{Xtalk}$$

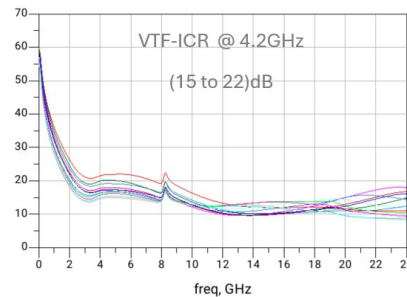
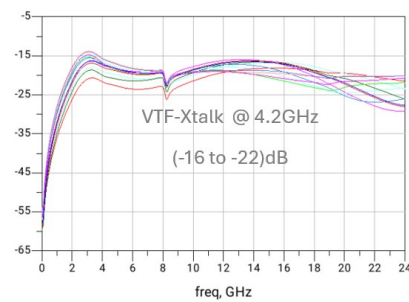
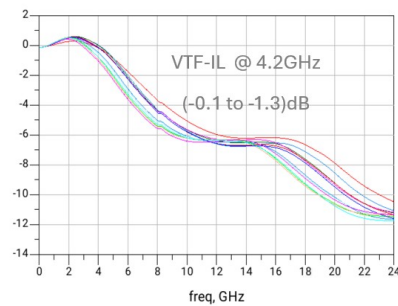


Fig2: VTF setup & plots: Conventional Si-Interposer for HBM3-8.4Gbps

Fig 2 illustrates the ADS setup for generating VTF metrics (Insertion Loss, Crosstalk, ICR) and presents the VTF-based channel characteristics of a conventional HBM3 channel using Si-Interposer, with the layout routing shown in Fig 1. The 4.2 GHz Nyquist frequency was selected for initial feasibility analysis at 8.4 Gbps, with plans to extend to 12.8 Gbps.

VTF plots offer the advantage of evaluating channel performance with actual Tx and Rx terminations included. S-parameter models of interposer were extracted at 25°C using a worst-case RC tech file from the foundry, using 3D EM tools (Clarity 3D/HFSS/Raptor).

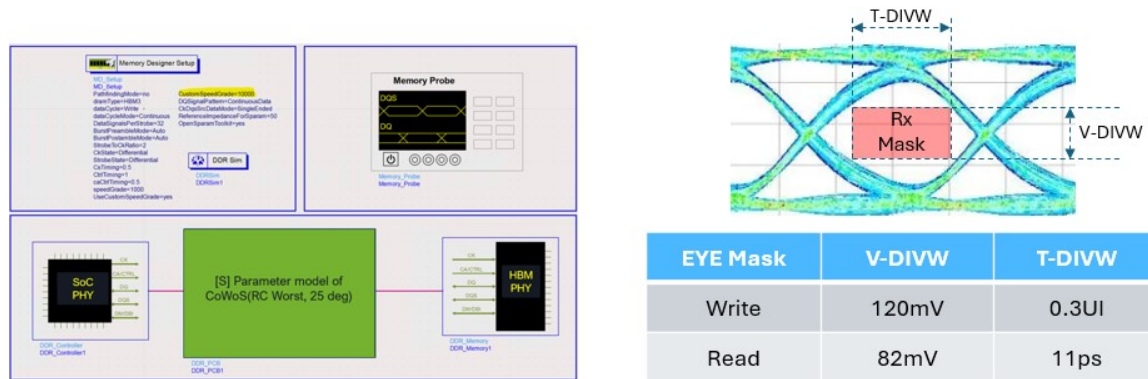


Fig 3, ADS-Memory designer setup and EYE Mask requirement

We began our feasibility analysis at 8.4Gbps, focusing on the SS corner IBIS model, which showed the worst-case eye opening. Thus, we present only SS corner results. Fig. 4 shows the simulated eye diagrams for read and write modes, generated using Memory Designer (Fig. 3). The SoC-PHY driver strength is set to 16mA, and the HBM-Memory driver strength is 14mA.

All aggressors were enabled and toggled with different PRBS8 pattern seeds. Statistical mode simulations extrapolated the eye contour for a BER of 1E-16. SoC-PHY and HBM-Memory Tx and Rx impairments like PSIJ, RJ, and DCD were excluded from the ADS simulation to isolate the interposer's contribution to eye closure and jitter, aiding in optimizing the interposer layout. In the subsequent jitter budget analysis, these impairments were included to assess the overall margin required to achieve 12.8Gbps.

Fig. 3 shows EYE mask requirements for HBM3 and HBM4 up to 12.8Gbps based on HBM3-6.4Gbps specs. With no JEDEC eye mask requirements for higher data rates, we used the HBM3 6.4Gbps requirement for write mode and our internal Alphawave SoC-PHY specification of 82mV eye height for read mode. We anticipate these requirements will be relaxed at 10Gbps and 12.8Gbps.

Fig. 4 shows that the write mode EYE width with a 120mV aperture mask is 40ps for the Si-Interposer conventional layout. For read mode, with an 82mV aperture mask, the EYE width is 58.93ps. These values indicate that the conventional layout does not meet the 8.4Gbps speed requirement in the system jitter budget for HBM3-8.4Gbps (Fig. 4).

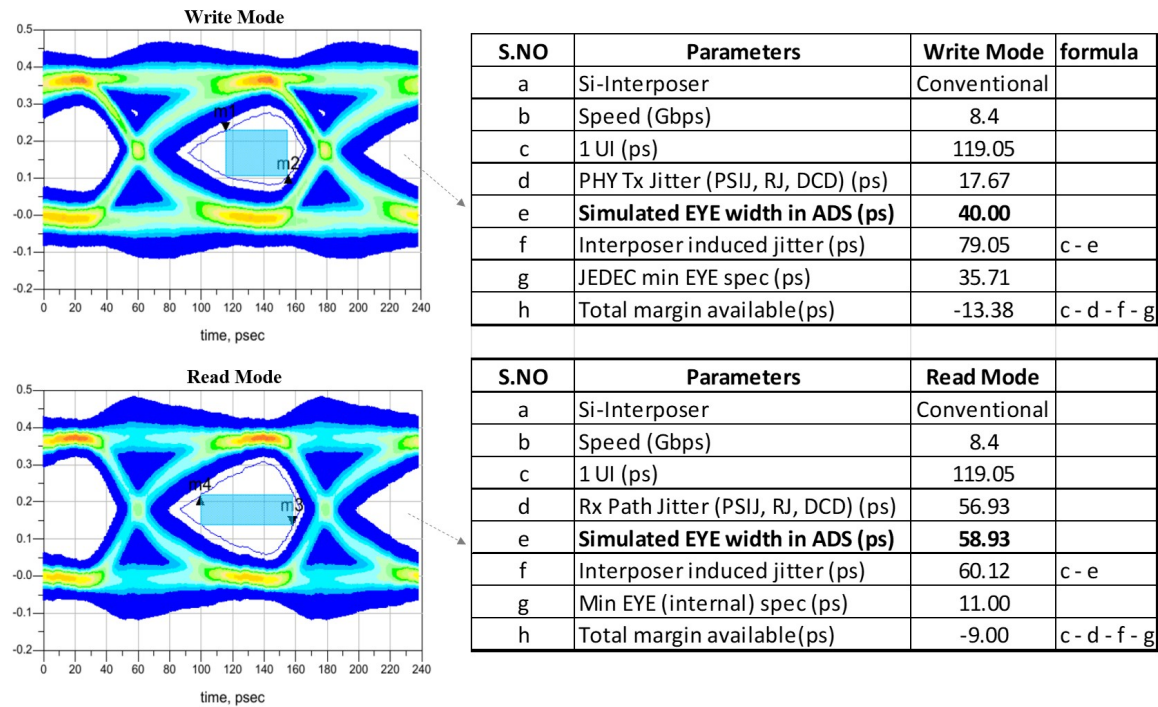


Fig 4: System jitter budget, HBM3-8.4Gbps, Conventional routing, Si-Interposer

1.2 Decomposing Interposer Induced Jitter

To optimize interposer layout and I/O architecture, it's essential to identify the dominant jitter type. We quantified ISI, crosstalk, and rise-fall degradation jitter by comparing eye widths under three conditions:

- 1. PRBS8 with Aggressors:** Eye width of the victim line toggling PRBS8 patterns with all aggressors active. This condition includes ISI, crosstalk, and rise-fall degradation.
- 2. Clock Pattern without Aggressors:** Eye width of the victim line toggling a clock pattern with aggressors off. This condition excludes ISI and crosstalk, isolating rise-fall degradation.
- 3. PRBS8 without Aggressors:** Eye width of the victim line toggling PRBS8 patterns with aggressors off. This condition excludes crosstalk but includes ISI and rise-fall degradation.

Using these measurements, we approximated the contributions of ISI, rise-fall degradation, and crosstalk to eye closure (refer to Fig 5 for calculations). For rise-fall degradation, the eye width with a clock pattern and no aggressors is 93.5ps, leaving a degradation of 25.55ps from a total UI width of 119.05 ps. ISI is calculated as the difference between the eye width with a clock pattern without aggressors (93.5ps) and with a PRBS8 pattern without aggressors (87.5ps), resulting in 6 ps. Crosstalk is determined by the total eye width loss with PRBS8 and aggressors, which is 79.05ps (119.05ps - 40ps). After subtracting the rise-fall degradation (25.55ps) and ISI (6ps), the crosstalk contribution is 47.5ps (79.05ps - 25.55ps - 6ps).

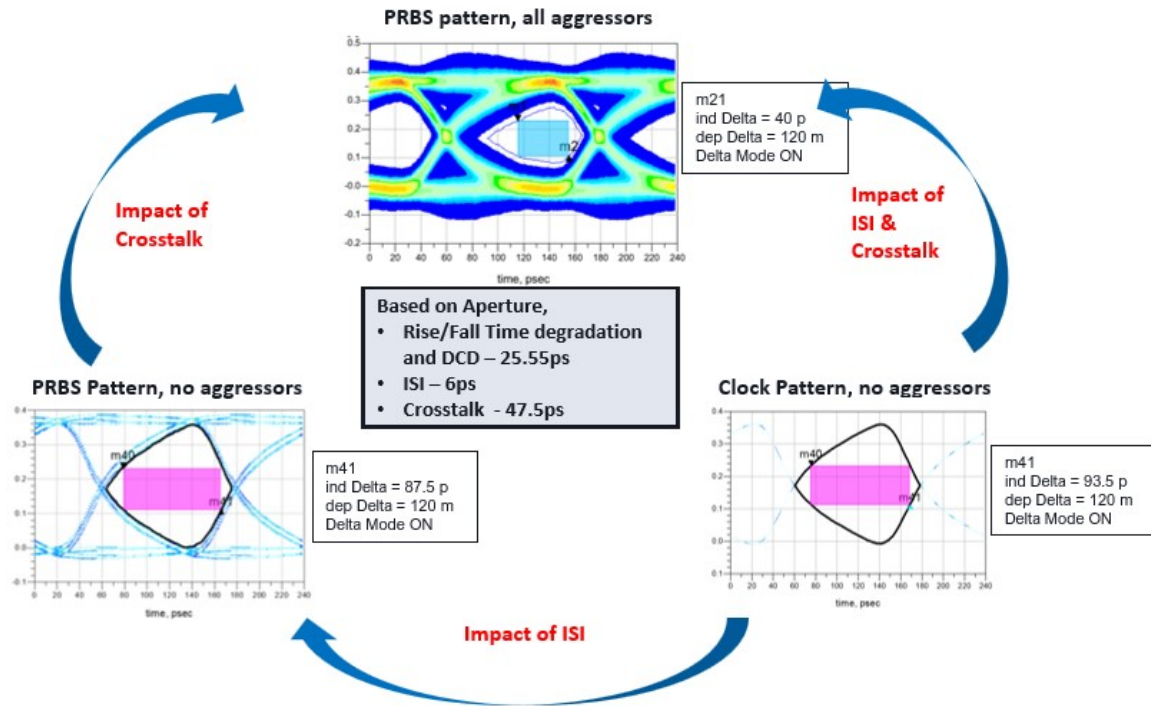


Fig 5, Interposer-Induced Jitter, Conventional routing, HBM3-8.4Gbps – write mode

	ISI	Crosstalk	Rise-fall degradation
Write	6ps	47.5ps	25.55ps
Read	4.76ps	35.12ps	20.24ps

Table 3, Interposer-Induced Jitter, Conventional, Si-Interposer, HBM3-8.4Gbps

Table 3 indicates that crosstalk is the primary factor for eye closure in Si-Interposer interposer channels, followed by rise-fall degradation due to channel loss. Increasing signal spacing was impractical due to the 378 μ m shoreline width constraint for fanning out 96 I/O ports in HBM3. Reducing signal trace width would increase insertion loss. Using three metal layers for signal routing, instead of two, would increase layer-to-layer crosstalk.

1.3 Novel Shield Structure to Reduce Crosstalk

To tackle these challenges, we developed a Novel [1] Si-Interposer signal-ground routing pattern, applicable for any number of metal layers. In this design, we used it for three metal signal layers, incorporating a shielding structure with ground grids and stitching vias (Fig-6). This design adheres to Si-Interposer layout constraints set by fabrication vendors. Each signal is enclosed by horizontal and vertical ground traces and stitching vias, significantly reducing crosstalk.

Optimizing the density of ground grids and stitching vias is critical: higher density reduces crosstalk but increases reflection and insertion loss. We created multiple test structures with varying densities of ground stitching vias and grids, extracting S-

Parameter models using 3D EM tools. By analyzing VTF metrics such as insertion loss (VTF-IL), power sum crosstalk (VTF-PSXT), and insertion loss to crosstalk ratio (VTF-ICR), we identified the optimal layout based on VTF-ICR. In the Si-Interposer stack-up, the dielectric thickness is very small. Aligning a ground trace directly beneath a signal trace increases capacitance and reflection loss. To mitigate this, we aligned signal routing over holes in the ground grid of adjacent layers, reducing capacitance while maintaining crosstalk shielding.

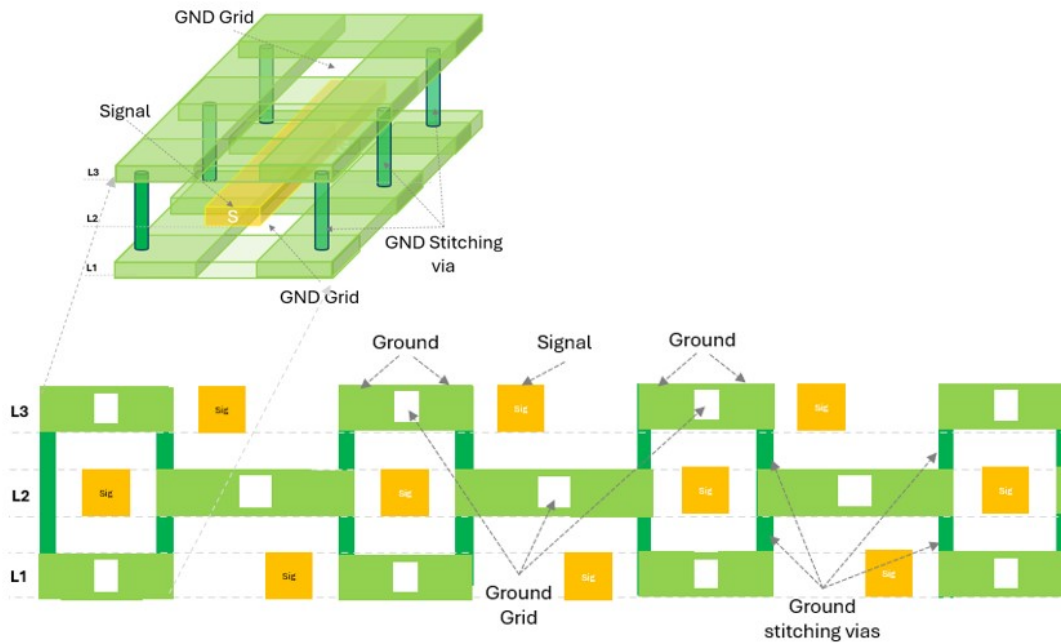


Fig 6, Novel shield structure for crosstalk reduction in Si-Interposer

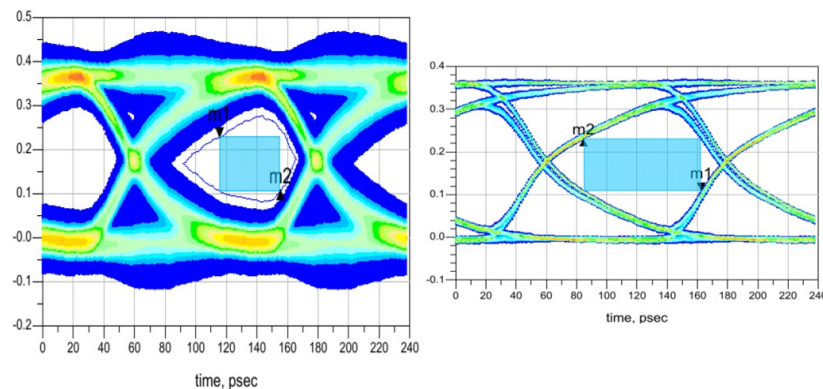


Fig 7: HBM3 8.4Gbps EYE - Conventional vs. Novel, Si-Interposer

Fig 8 compares the VTF channel characteristics of conventional and Novel layout. Despite increased insertion loss, the Novel layout significantly reduced crosstalk by ~18 dB, resulting in a ~18 dB increases in VTF-ICR, as reflected in the eye improvement shown in Fig. 7.

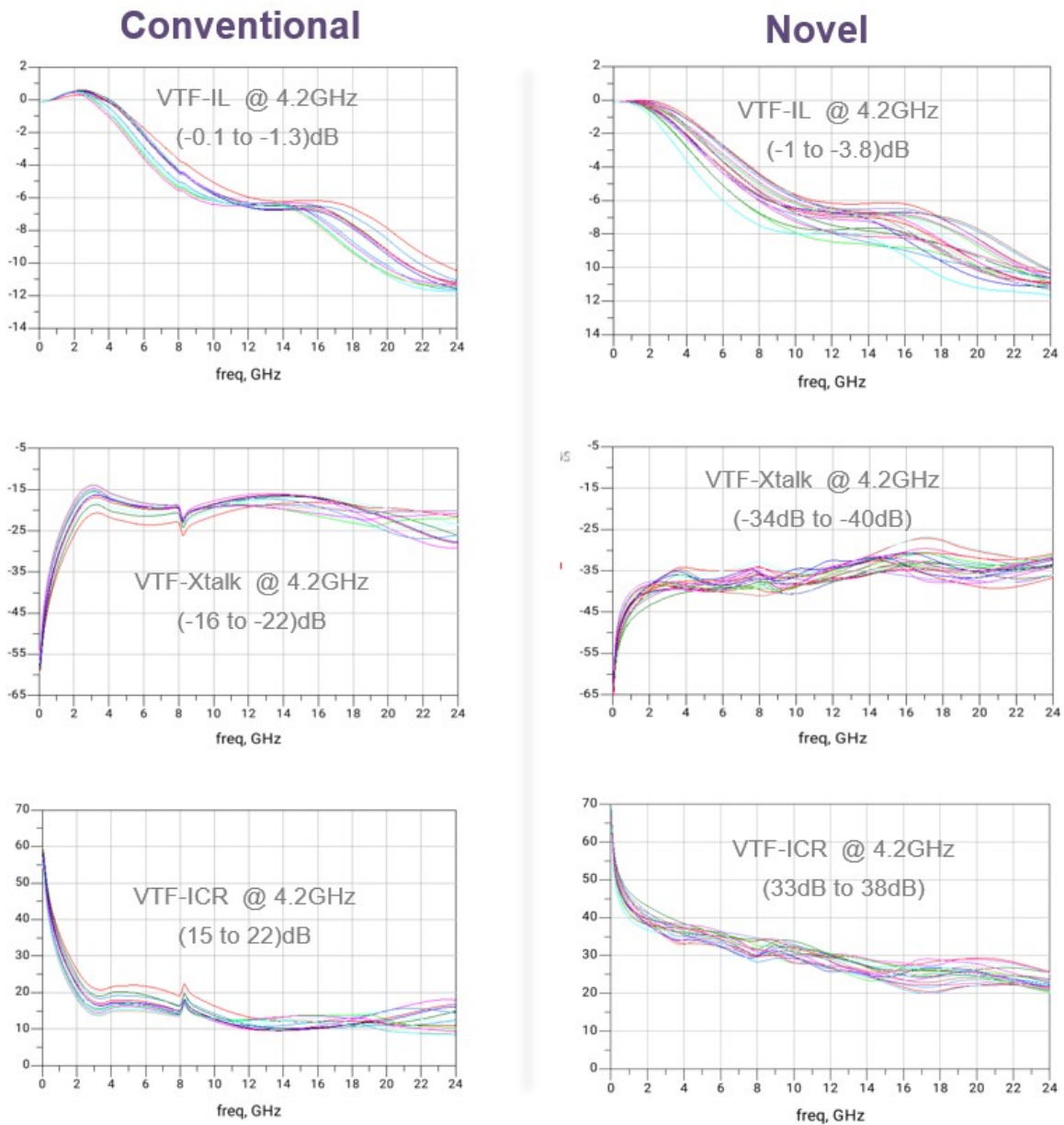


Fig 8, VTF based channel parameters - Conventional Vs Novel

Table 4 compares the jitter budget for 8.4Gbps between conventional and Novel layouts. The conventional layout fails to meet the 8.4Gbps requirement, while the Novel layout comfortably achieves it. Although the Novel layout meets the 9.6Gbps requirement, the margin is minimal. It may fail with an interposer model at 105°C (current models are at 25°C) and is likely to fail at 12.8Gbps. Thus, further optimization is needed to meet the HBM3-12.8Gbps requirement.

HBM3 - Write Mode					
S.NO	Parameters	8.4Gbps		9.6Gbps	formula
a	Si-Interposer layout pattern	Conventional	Novel (3L)	Novel (3L)	
b	Speed (Gbps)	8.4	8.4	9.6	
c	1 UI (ps)	119.05	119.05	104.17	
d	PHY Tx Jitter (PSIJ, RJ, DCD) (ps)	17.67	17.67	17.67	
e	Simulated EYE width in ADS (ps)	40.00	81.00	55.00	
f	Interposer induced jitter (ps)	79.05	38.05	49.17	c - e
g	JEDEC min EYE spec (ps)	35.71	35.71	31.25	
h	Total margin available(ps)	-13.38	27.62	6.08	c - d - f - g

HBM3 - Read Mode					
S.NO	Parameters	8.4Gbps		9.6Gbps	formula
a	Si-Interposer layout pattern	Conventional	Novel (3L)	Novel (3L)	
b	Speed (Gbps)	8.4	8.4	9.6	
c	1 UI (ps)	119.05	119.05	104.17	
d	Rx Path Jitter (PSIJ, RJ, DCD) (ps)	56.93	56.93	51.21	
e	Simulated EYE width in ADS (ps)	58.93	82.14	66.00	
f	Interposer induced jitter (ps)	60.12	36.91	38.17	c - e
g	Min EYE (internal) spec (ps)	11.00	11.00	11.00	
h	Total margin available(ps)	-9.00	14.21	3.79	c - d - f - g

Table 4: Jitter budget - Conventional vs. Novel Si-Interposer layout

To understand the interposer-induced jitter components of the Novel layout, we decomposed the jitter into ISI, rise-fall degradation, and crosstalk. Table 5 shows that while the Novel layout significantly reduces crosstalk, it increases rise-fall degradation due to a ~1.5dB increase in channel loss. To reach 12.8Gbps, we need to reduce this loss and find a balance between crosstalk and loss.

HBM3-8.4Gbps						
	ISI		Crosstalk		Rise-fall degradation	
Write	6ps	3ps	47.5ps	1.72ps	25.55ps	33.35ps
Read	4.76ps	5.36ps	35.12ps	2.98ps	20.24ps	28.57ps

Table 5: Interposer induced jitter, Conventional Vs. Novel Si-Interposer layout

2.0 Novel shield structure with 4 layers for signal

To address insertion loss without increasing crosstalk, we opted for a four-layer signal configuration. Based on our experience with Si-Interposer, we adopted the routing pattern shown in Fig-9 (c). It is similar to our novel shield and routing pattern but uses four signal layers [10]. The trace width and spacing are optimized accordingly. To minimize capacitance, signals are not placed directly on top of ground traces. Each signal is shielded by six ground traces, ground grids and ground stitching vias to reduce crosstalk. The variables for optimization, marked in Fig-9 (a), include Signal Trace width (A), Ground Trace width (B), Signal to Ground spacing (C), and Ground to Ground spacing (D). We need to sweep these variables to determine the optimal layout, but the model extraction time poses a significant bottleneck.

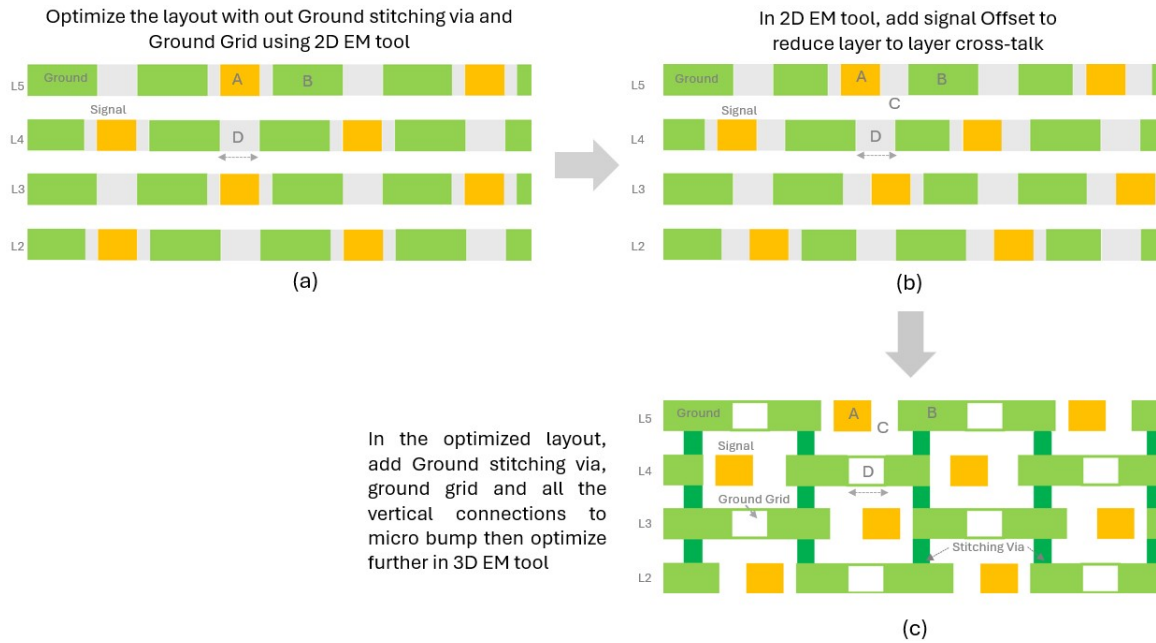


Fig-9: Si-Interposer routing pattern with 4 layer assigned for signal routing

2.1 Efficient Layout Optimization Methodology

To optimize interposer design efficiently, we utilize both 2D and 3D EM tools to streamline parameter sweeps and reduce extraction times. Conventional 3D EM tools like HFSS, Raptor, and Clarity 3D can take 8 to 12 hours to extract a 6–8-layer interposer stack-up, making rapid layout parameter adjustments challenging. To accelerate this, we begin with Ansys’s 2D extraction tool for initial sweeps of critical layout parameters—such as signal trace width, ground trace width, and spacing—as shown in Fig. 9a. This approach allows for quick analysis of layout variations and their impact on channel characteristics, enabling optimization based on key metrics like VTF-based Insertion Loss (IL), Power Sum Crosstalk (PSXT), and Insertion Loss to Crosstalk Ratio (ICR) following our SI-PI methodology.

Fig. 10 demonstrates how layout parameters (depicted in Fig. 9a) affect channel metrics (IL, PSXT, ICR). For instance, increasing the ground trace width reduces crosstalk but raises both return and insertion loss, while widening the ground trace spacing (D) increases crosstalk. Case 8, which maximizes VTF-ICR, was chosen for further refinement. To minimize layer-to-layer crosstalk, a calculated offset was applied to signal alignments (fig 9b) between layers L2/L4 and L3/L5 using the 2D EM tool.

This optimized layout was then incorporated into the full design, which includes micro bumps, a ground grid structure (featuring ground stitching vias and horizontal/vertical ground traces), and via connections, as shown in Fig. 9c. The final model was extracted using a 3D EM tool, with an optimized density of ground grid and stitching vias to balance extraction accuracy and simulation speed. This model will be used for system-level EYE simulation with IBIS models for SoC-PHY and HBM-PHY using ADS Memory Designer (refer to Fig. 3).

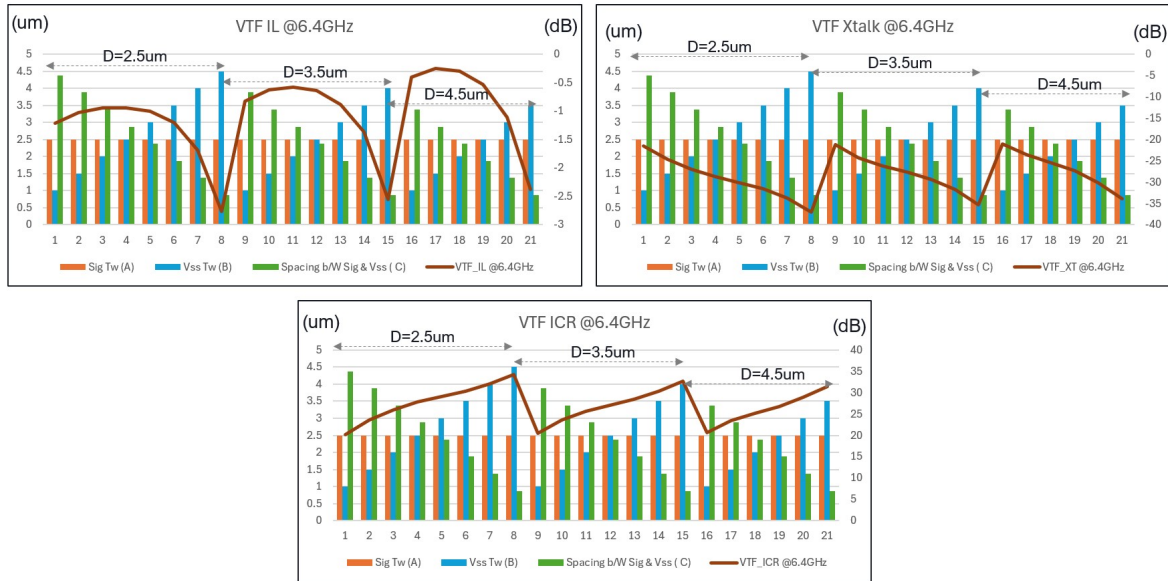


Fig-10: Layout Parameters Vs. Channel Parameters

HBM3 - Write Mode			
S.NO	Parameters	12.8Gbps	formula
a	Si-Interposer layout pattern	Novel (4L)	
b	Speed (Gbps)	12.8	
c	1 UI (ps)	78.13	
d	PHY Tx Jitter (PSIJ, RJ, DCD) (ps)	17.67	
e	Simulated EYE width in ADS (ps)	47.27	
f	Interposer induced jitter (ps)	30.86	c - e
g	JEDEC min EYE spec (ps)	23.44	
h	Total margin available(ps)	6.16	c - d - f - g
HBM3 - Read Mode			
S.NO	Parameters	12.8Gbps	formula
a	Si-Interposer layout pattern	Novel (4L)	
b	Speed (Gbps)	12.8	
c	1 UI (ps)	78.13	
d	Rx Path Jitter (PSIJ, RJ, DCD) (ps)	44.65	
e	Simulated EYE width in ADS (ps)	55.47	
f	Interposer induced jitter (ps)	22.66	c - e
g	Min EYE (internal) spec (ps)	11.00	
h	Total margin available(ps)	-0.18	c - d - f - g

Table 6, HBM3-12.8Gbps Jitter budget for Novel-4L Si-Interposer

Table 6 shows the jitter budgets for HBM3 at 12.8 Gbps. The 4L signal with novel shielding meets the 12.8 Gbps target, though with limited margin. To increase this margin, we should explore equalization options in IO design, which will be discussed later in the paper.

Please Note: we are currently using a 120-mV rectangular mask for Write and 82 mV for Read, suitable for lower speeds. However, as speeds increase, hexagonal EYE masks with an EYE height requirement of 80 mV or less may be introduced. Adopting such masks would significantly improve margin, making 12.8 Gbps achievable with the 4-layer signal design in Si-Interposer when paired with the optimized novel shielding described in this paper.

Power integrity remains a key factor, as increasing signal layers can reduce power layers and potentially elevate IR drop. A detailed EMIR analysis would be necessary, though this is outside the scope of this paper.

3.0 Transitioning from HBM3 to HBM4

In HBM4, Interface width have doubled to 2048 compared to HBM3's 1024. These ports are organized into independent channels, with HBM3 supporting 16 channels and HBM4 expanding to 32. Fig 10 illustrates the micro bump pin arrangements for both.

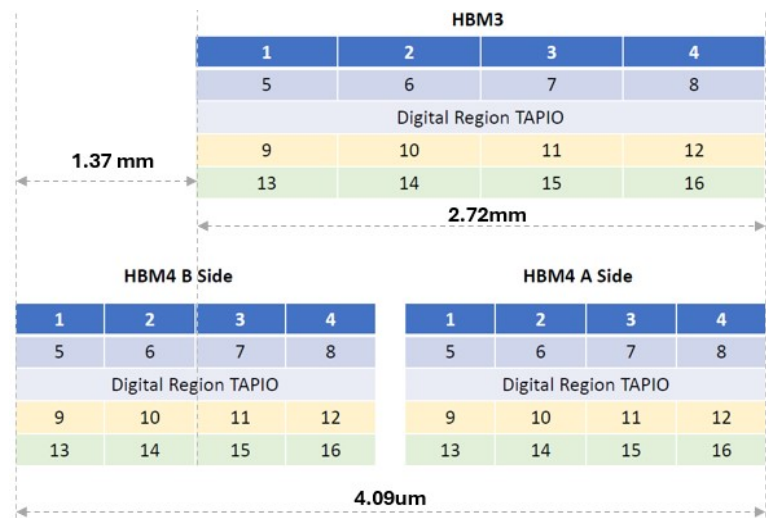


Fig 10: Comparison of Micro Bump Pin Field Length: HBM3 vs. HBM4

For our HBM4 case study, we maintained the same shoreline width as HBM3 and used a 48x55 Micro Bump pitch for HBM-PHY. HBM4's interposer routing length increased by ~2 mm over HBM3, with worst-case lengths at ~5.1 mm for HBM3 and ~7.1 mm for HBM4, resulting in higher signal loss. Additionally, doubling the signals within the same shoreline width amplifies crosstalk, complicating the goal of achieving 12.8 Gbps.

Given these signal integrity challenges with HBM4, we opted to explore Organic Interposer instead of Si-Interposer to meet the 12.8 Gbps target. While Si-Interposer has shown marginal success at 12.8 Gbps with HBM3 (Table 6), Organic Interposer offers low RC interconnect, less insertion loss channels [2], and less EYE rise-fall degradation, making it more suitable for HBM4 at 12.8 Gbps.

Fig 11a depicts a section of the HBM4 micro bump pin map in the SoC PHY region, featuring 24 data I/O surrounded by power and ground micro bump pins. HBM4 has 8 such sections stacked horizontally, resulting in 192 signals (8 x 24) within a 380 μm width, a subset of the total 2048 I/O ports. Signal pitch is 7.9 μm with 4 layers for signal routing and 11.9 μm with 6 layers for signal routing (Fig 11d). This signal pitch accommodates the signal trace width, ground trace width, and spacing between signal and ground traces.

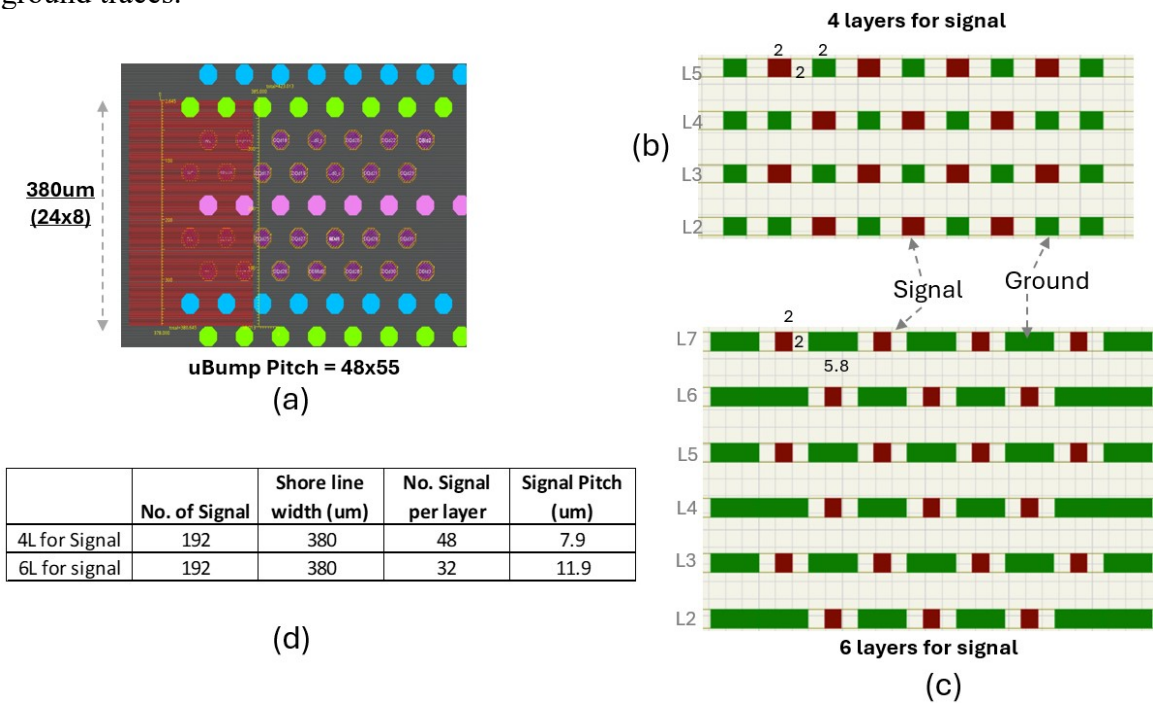


Fig11: Signal pitch calculation for Organic Interposer (4 and 8 layers for signal)

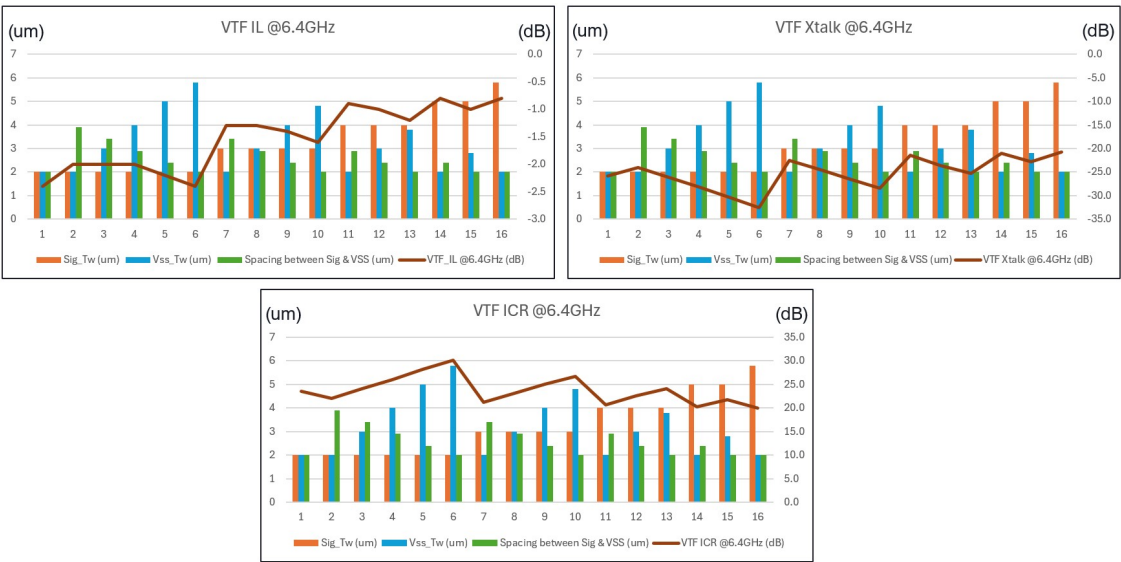


Fig 12: Optimizing channel parameters for Organic Interposer

3.1 Optimized layout for Organic Interposer

Based on the layout optimization exercise shown in Fig 12, Fig. 11 illustrates the optimized G-S-G pattern for Organic Interposer with 4 and 6 signal routing layers respectively. Unlike Si-Interposer, Organic Interposer features thicker dielectric layers. Aligning signal and ground traces in Organic Interposer reduces layer-to-layer crosstalk and optimizes performance.

In Si-Interposer, thinner dielectric layers increase capacitance, raising Return and Insertion losses. Fig 12 shows insertion loss is less critical than crosstalk. Increasing signal trace width can reduce insertion loss but increases crosstalk by narrowing ground traces, lowering ICR. Thus, minimizing crosstalk is more crucial than reducing insertion loss in both Si-Interposer and Organic Interposer.

The optimized layout from Fig12 was implemented, and [S] parameters were extracted using a 3D-EM tool. Table 7 compares the channel parameters of Organic Interposer with 4 and 6 signal routing layers. With 4 layers, the ICR achieved was 23.6 dB at 6.4 GHz, while 6 layers achieved 32.9dB, an improvement of ~9.3dB. This model will be used for system-level EYE simulation with IBIS models for SoC-PHY and HBM-PHY using ADS Memory Designer (refer to Fig. 3).

Organic Interposer	VTF IL @6.4GHz (dB)	VTF Xtalk @6.4GHz (dB)	VTF ICR @6.4GHz (dB)
4L for Signal	-4.1dB	-27.7dB	23.6dB
6L for Signal	-3.4dB	-36.3dB	32.9dB

Table 7, Optimized channel parameters of Organic Interposer

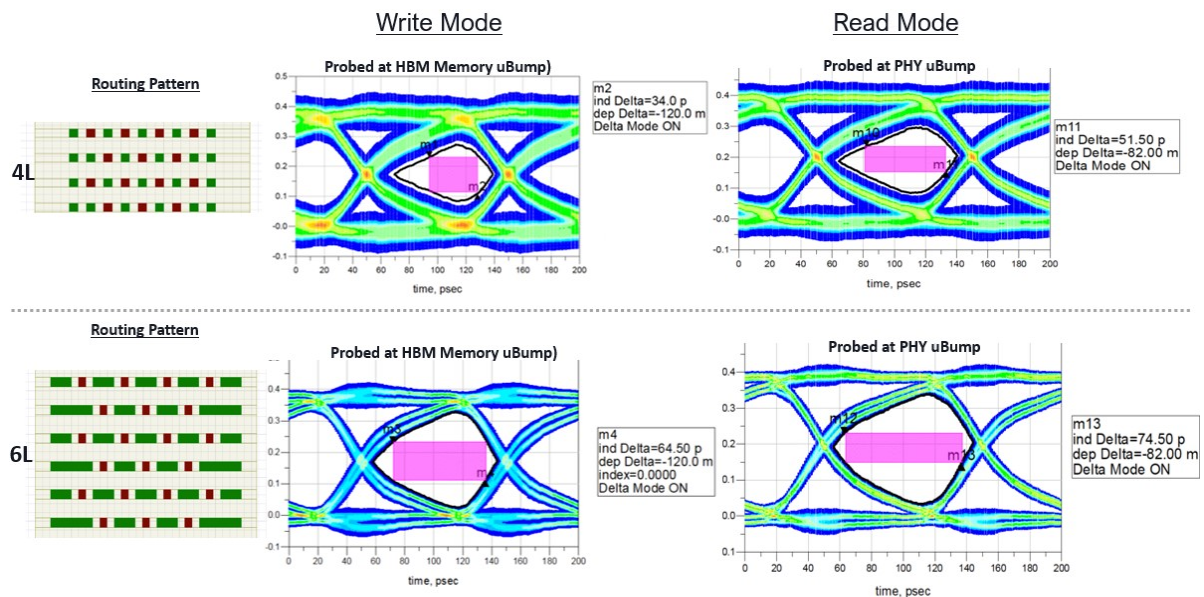


Fig 13, EYE for HBM4-10Gbps, Organic Interposer, 4L vs. 6L for signal.

3.2 Organic Interposer Induced Jitter - HBM4 10Gbps

In Fig 13, the available write mode EYE width with a 120mV aperture mask is 34ps for Organic Interposer (4L for signal) 60.5ps for Organic Interposer (6L for signal). For read mode, with an 82mV aperture mask, the EYE width is 51.5ps for Organic Interposer (4L for signal) and 74.5ps for Organic Interposer (6L for signal). According to the system jitter budget for HBM4-10Gbps (Table 8), Organic Interposer (4L for signal) fails to meet the 10Gbps requirement, while Organic Interposer (6L for signal) provides a comfortable margin.

S.NO	Parameters	Organic, Write Mode - HBM4 10Gbps		formula
a	Number of layer assigned for signal	4L	6L	
b	Speed (Gbps)	10	10	
c	1 UI (ps)	100.00	100.00	
d	PHY Tx Jitter (PSIJ, RJ, DCD) (ps)	17.67	17.67	
e	Simulated EYE width in ADS (ps)	34.00	64.50	
f	Interposer induced jitter (ps)	66.00	35.50	c - e
g	JEDEC min EYE spec (ps)	30.00	30.00	
h	Total margin available(ps)	-13.67	16.83	c - d - f - g
S.NO	Parameters	Organic, Read Mode - HBM4 10Gbps		
a	Number of layer assigned for signal	4L	6L	
b	Speed (Gbps)	10	10	
c	1 UI (ps)	100.00	100.00	
d	Rx Path Jitter (PSIJ, RJ, DCD) (ps)	51.21	51.21	
e	Simulated EYE width in ADS (ps)	51.50	74.50	
f	Interposer induced jitter (ps)	48.50	25.50	c - e
g	Min EYE (internal) spec (ps)	11.00	11.00	
h	Total margin available(ps)	-10.71	12.29	c - d - f - g

Table 8, Jitter budget sheet for HBM4 - 10Gbps

HBM4-10Gbps	ISI	Crosstalk	Rise-fall degradation
4L for signal	7ps	33ps	26ps
6L for signal	7ps	6.5ps	22ps

Table 9, Interposer-induced jitter: Organic Interposer (4L vs 6L for signal), 10Gbps.

To further optimize for 12.8Gbps, we decomposed interposer-induced jitter into ISI, crosstalk, and rise-fall degradation. Figs. 14 and 15 show the jitter decomposition for Organic Interposer (4L for signal) and Organic Interposer (6L for signal), respectively. Table 9 compares interposer-induced jitter between these configurations. In Organic Interposer (4L for signal), crosstalk is the main source of interposer-induced jitter, followed by rise-fall degradation and ISI. Organic Interposer (6L for signal) reduces crosstalk significantly due to wider ground traces and increased signal spacing. Further optimizing Organic Interposer (4L for signal) is challenging because the signal pitch is only 7.9 μm with a minimum trace width/spacing of 2 μm [2], leaving no room to reduce

crosstalk. Addressing rise-fall degradation by increasing signal trace width would worsen crosstalk. To achieve 10Gbps with Organic Interposer (4L for signal), consider enhancing I/O design with equalization options like CTLE, 1/2-tap DFE to address rise-fall degradation and ISI. Further discussion on equalization will be deferred until exploring options for 12.8Gbps.

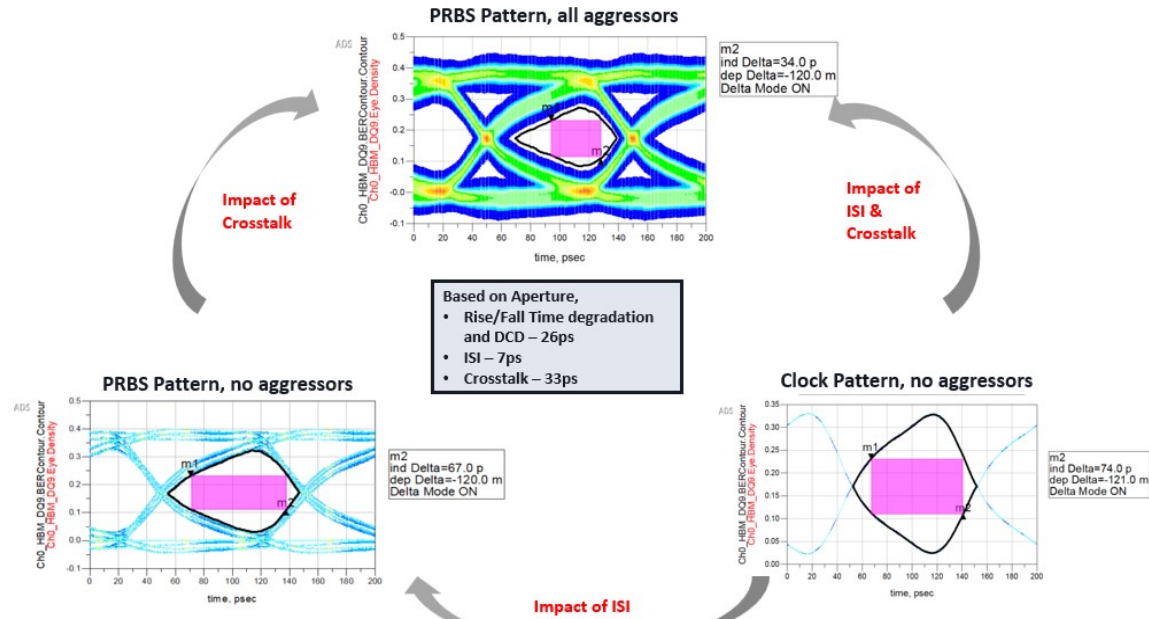


Fig 14, Organic Interposer (4L for signal), interposer-induced jitter – 10Gbps.

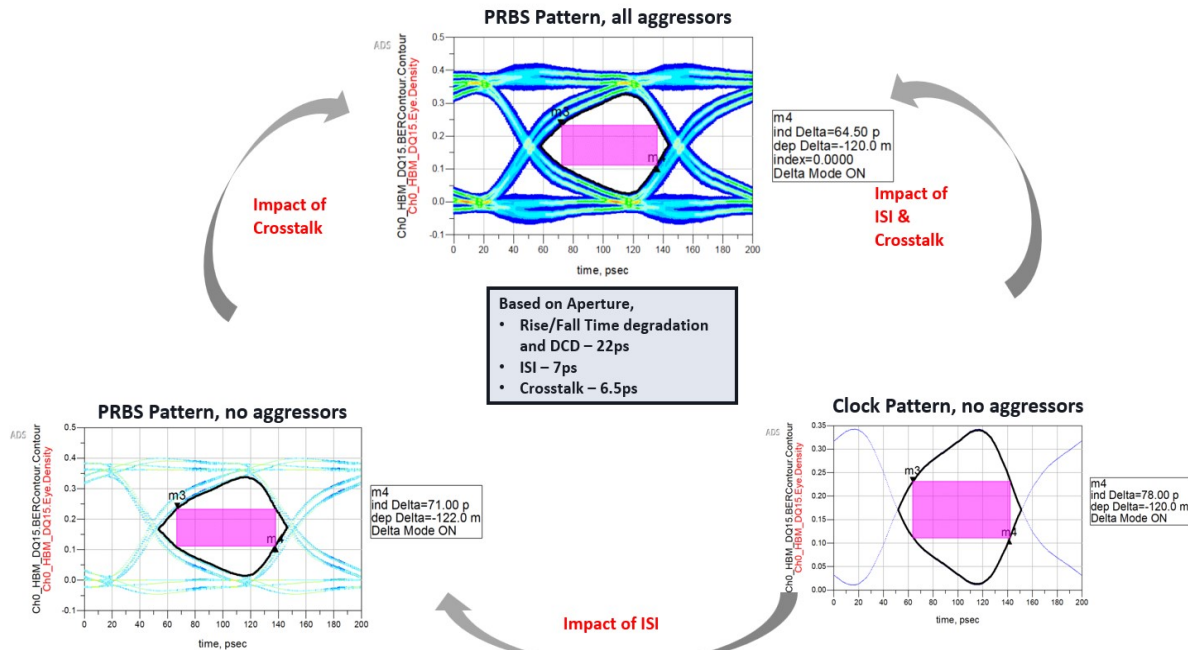


Fig 15, Organic Interposer (6L for signal), interposer-induced jitter – 10Gbps

3.4 Feasibility of Organic Interposer for HBM4 at 12.8 Gbps

Fig 16 shows interposer-induced jitter for 12.8 Gbps with Organic Interposer with 6L for signal. Crosstalk and ISI are less significant compared to rise-fall degradation. Table 11 provides the jitter budget for HBM4 at 12.8 Gbps. Without equalization ("No EQ" column), the design doesn't meet the 12.8 Gbps requirement due to rise-fall degradation from channel loss, though crosstalk is controlled.

Efforts to optimize the layout by increasing signal trace width and decreasing ground shielding width yielded minimal insertion loss reduction but increased return loss and crosstalk, reducing ICR and EYE margin (see Table 10 and Fig 17). The optimal layout for Organic Interposer with 6L for signal is Case 1, with a 2 μm signal trace width, 2 μm spacing, and 5.8 μm ground trace width. To address interposer-induced jitter from rise-fall degradation and ISI, exploring Tx and Rx equalization is necessary.

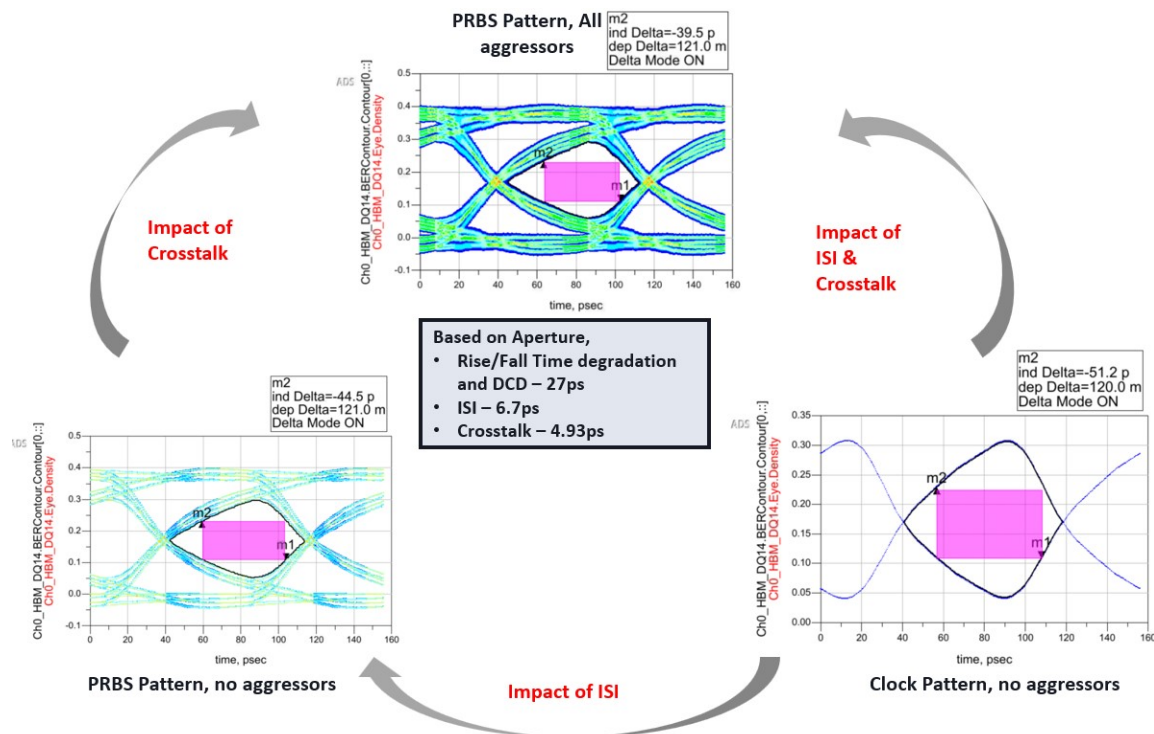


Fig 16, Organic Interposer(6L for signal) interposer-induced jitter – 12.8Gbps

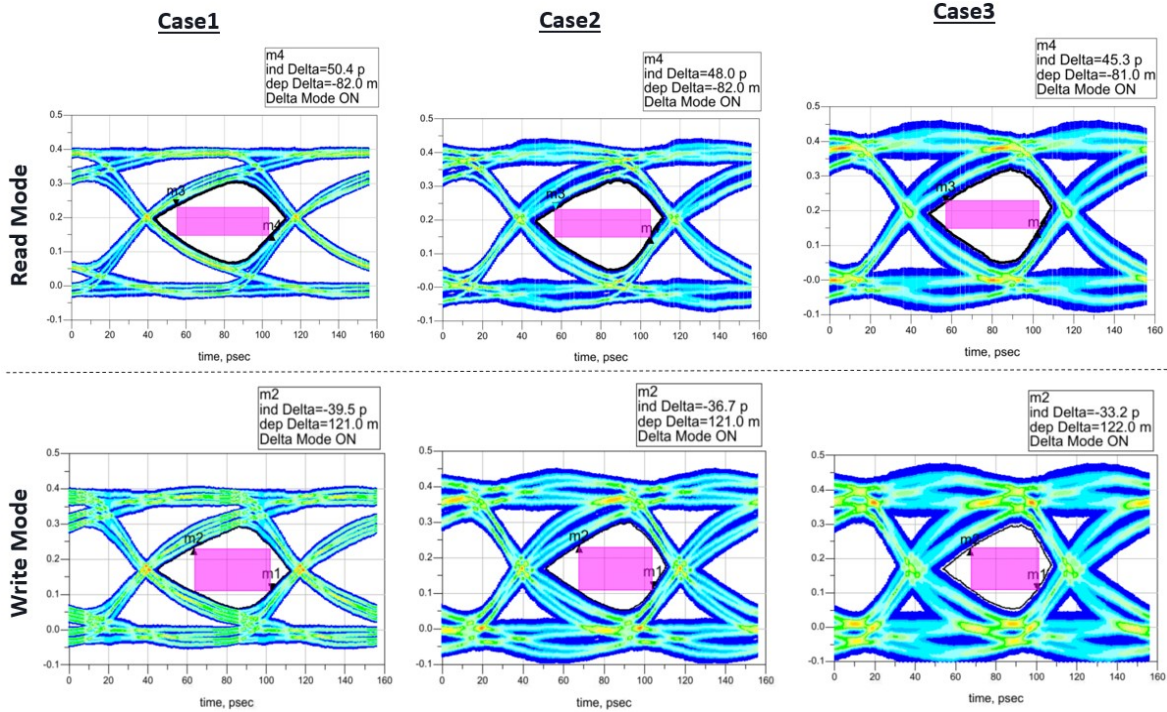


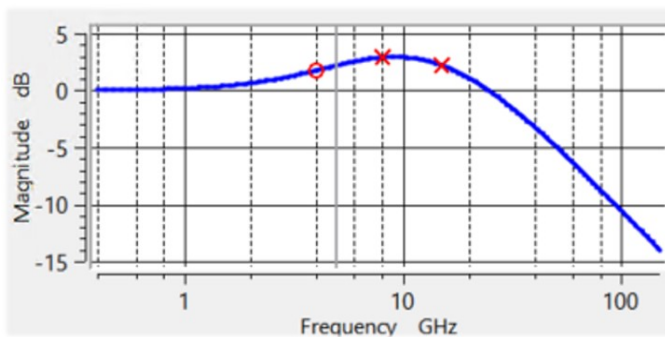
Fig 17, Signal trace width Vs. EYE

Sig_Tw	VSS_Tw	VTF IL @6.4GHz (dB)	VTF Xtalk @6.4GHz (dB)	VTF ICR @6.4GHz (dB)
2um	5.8um	-3.4dB	-36.3dB	32.9dB
3um	4.8um	-2.4dB	-31.3dB	28.9dB
4um	3.8um	-1.75dB	-27.1dB	25.35dB

Table 10, Signal trace width Vs. ICR

3.5 Explore Equalization to achieve 12.8 Gbps.

As seen in Fig 16, the design is limited by rise-fall degradation due to high frequency channel loss. To address this, a high-frequency boost with a carefully selected peaking frequency (e.g. CTLE), selected based on signal's rise and fall time, which determines its high-frequency content.



$$H(s) = \frac{A_{dc}\omega_{p1}\omega_{p2}}{\omega_z} \cdot \frac{s + \omega_z}{(s + \omega_{p1})(s + \omega_{p2})}$$

Fig 18, 2 pole 1 zero CTLE with peaking at 10GHz

After several experiments with CTLE curves, we opted for a 2-pole, 1-zero CTLE configuration. This configuration includes a zero frequency at 4 GHz, pole1 at 8 GHz, and pole2 at 15 GHz, resulting in approximately 3 dB peaking at 10 GHz (see Fig 18).

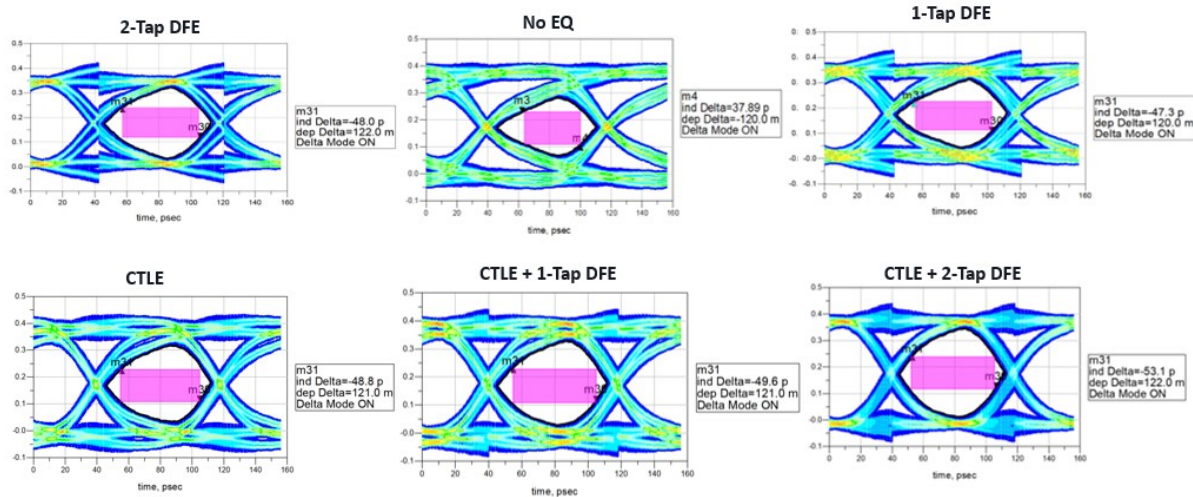


Fig 19a, Effect of CTLE and 1 & 2 tap DFE on HBM4 12.8GBps – Write Mode

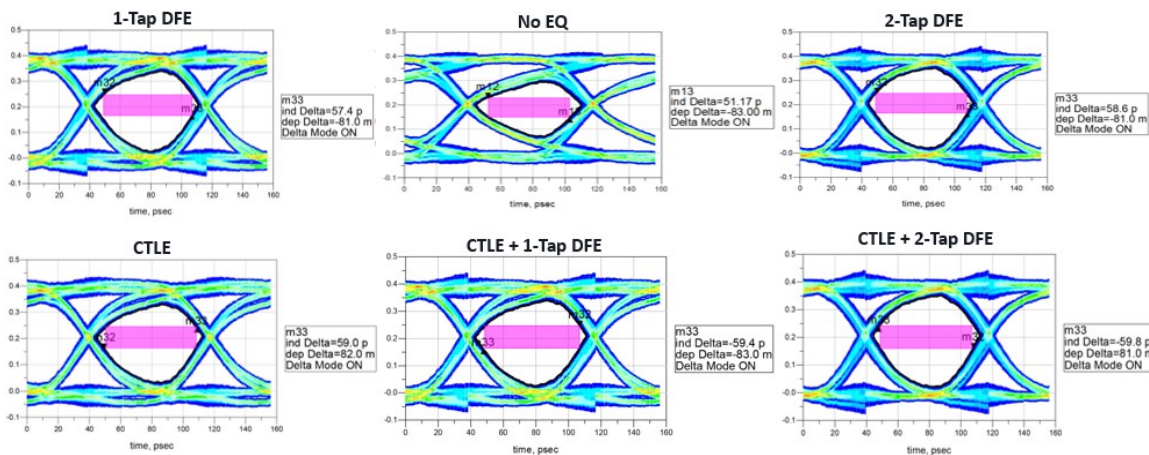


Fig 19b, Effect of CTLE and 1 & 2 tap DFE on HBM4 12.8GBps – Read Mode

S.NO	Parameters	Write Mode - HBM4 12.8Gbps						formula
a	Equalizaion	No EQ	CTLE	1 tap DFE	CTLE + 1 tap DFE	2 tap DFE	CTLE + 2 tap DFE	
b	Speed (Gbps)	12.8	12.8	12.8	12.8	12.8	12.8	
c	1 UI (ps)	78.13	78.13	78.13	78.13	78.13	78.13	
d	PHY Tx Jitter (PSIJ, RJ, DCD) (ps)	17.67	17.67	17.67	17.67	17.67	17.67	
e	Simulated EYE width in ADS (ps)	39.50	48.80	47.30	49.60	48.00	53.10	
f	Interposer induced jitter (ps)	38.63	29.33	30.83	28.53	30.13	25.03	c - e
g	JEDEC min EYE spec (ps)	23.44	23.44	23.44	23.44	23.44	23.44	
h	Total margin available(ps)	-1.61	7.69	6.19	8.49	6.89	11.99	c - d - f - g

S.NO	Parameters	Read Mode - HBM4 12.8Gbps						
a	Equalizaion	No EQ	CTLE	1 tap DFE	CTLE + 1 tap DFE	2 tap DFE	CTLE + 2 tap DFE	
b	Speed (Gbps)	12.8	12.8	12.8	12.8	12.8	12.8	
c	1 UI (ps)	78.13	78.13	78.13	78.13	78.13	78.13	
d	Rx Path Jitter (PSIJ, RJ, DCD) (ps)	44.65	44.65	44.65	44.65	44.65	44.65	
e	Simulated EYE width in ADS (ps)	51.70	59.00	57.40	59.40	58.60	59.80	
f	Interposer induced jitter (ps)	26.43	19.13	20.73	18.73	19.53	18.33	c - e
g	Min EYE (internal) spec (ps)	11.00	11.00	11.00	11.00	11.00	11.00	
h	Total margin available(ps)	-3.95	3.35	1.75	3.75	2.95	4.15	c - d - f - g

Table 11, Jitter budget sheet for HBM4 – 12.8Gbps, write and read mode

Although this may not be the optimized CTLE for this channel, it serves as a proof of concept to demonstrate the potential improvement in eye opening. Figs 19a (write mode) and 19b (read mode) show the effects of CTLE, 1-tap DFE, 2-tap DFE, and CTLE combined with 1-tap/2-tap DFE. Table 11 summarizes the overall margin improvement for HBM4 at 12.8 Gbps in both write and read modes.

Note: In this paper, we use 120mV as vertical EYE mask requirements at 12.8 Gbps. However, these requirements may be relaxed to achieve higher speeds. For example, a hexagonal EYE mask with a lower height threshold may be incorporated into our optimization framework. Adopting these would significantly reduce the impact of rise-fall degradation and could eliminate the need for equalization.

4.0 Summary & Conclusion

This paper details our journey of integrating HBM Memory into SoC using Si & Organic Interposer technology, advancing from HBM3 at 7.2 Gbps to HBM4 at 12.8 Gbps, driven by AI boom. Key industry questions addressed include:

1. Achieving 12.8 Gbps Integration: Methods for integrating HBM4 and HBM3 memory to SoC using Si & Organic interposer technology.
2. Suitable Interposer Technology: Evaluating Si-Interposer, Organic Interposer for optimal performance.
3. Required Interposer Layers: Determining the necessary layers and their allocation for signals, power, and decoupling caps.
4. Characterizing Interposer Channels with VTF based (IL, PSXT, ICR) and EYE (at BER 1E-16) for full characterization.

5. Optimized Signal-Ground Routing pattern: Providing an optimized signal-ground routing pattern to meet 12.8 Gbps.
6. Novel Shield Structure: Introducing a Novel shield structure [1][10] to significantly reduce crosstalk.
7. Interposer-Induced Jitter: Assessing the impact of interposer-induced jitter on the system's jitter budget and decomposing it into ISI, crosstalk, and rise-fall time degradation to identify the dominant component.
8. Channel Parameters Affecting EYE Closure: Determining the dominant factors affecting EYE closure (crosstalk, channel loss, or reflection).
9. SI/PI Methodologies: Detailing methodologies to identify and address critical channel parameters in layout or I/O architecture.
10. Layout Optimization: Presenting a methodology for optimizing layout using 2D and 3D extraction tools, balancing between accuracy and extraction time.
11. I/O Architecture Enhancements: Evaluating the need for enhancements like equalization for 12.8 Gbps.

The study begins by analyzing conventional Si-Interposer signal-ground routing for HBM3, focusing on VTF based insertion loss, PSXT, and ICR. EYE simulations and jitter analysis show these layouts struggle above HBM3-7.2 Gbps, mainly due to crosstalk. To address this, we devised a Novel crosstalk shielding structure [1][10], which reduces crosstalk and enhances data rates up to 9.8 Gbps. However, further optimization is needed to achieve 12.8 Gbps for HBM3 and HBM4.

To optimize the layout, we need to understand interposer-induced jitter by decomposing it into ISI, crosstalk, and rise-fall time degradation. This analysis will identify key channel parameters affecting EYE closure, guiding layout, and I/O architecture improvements. Once identified, we can optimize the layout by systematically adjusting parameters like signal trace width, ground trace width, signal-to-ground spacing, alignment between signal and ground layers, and signal-ground routing patterns. We will study parameters like VTF (IL, PSXT, and ICR) for each scenario to select the optimal layout based on VTF-ICR.

The extraction process with 3D tools like HFSS, Raptor, and Clarity 3D is time-consuming, requiring 8 to 9 hours per case study. To expedite this, we use a 2D extraction tool from Ansys for quick parameter sweeps, followed by 3D EM tools for final S-parameter model extraction. This combination balances accuracy and speed, enabling efficient layout optimization.

Finally, we present the optimized Si & Organic interposer layouts for HBM3 and HBM4 at 12.8 Gbps, along with a system-level jitter budget sheet showing the available margin for 12.8 Gbps.

Further analysis for 12.8 Gbps reveals that, even with the optimized Si & Organic Interposer layout, interposer-induced jitter from rise-fall time degradation remains a limiting factor. This necessitates the exploration of equalization techniques. The study considers a 2-pole, 1-zero continuous-time linear equalizer (CTLE) and evaluates the effects of 1-tap and 2-tap decision feedback equalization (DFE) on the system performance. The combination of CTLE and DFE significantly improves the eye opening for both write and read modes, demonstrating the feasibility of achieving 12.8 Gbps for HBM4 with the optimized Organic Interposer layout and appropriate equalization techniques.

In conclusion, our study provides a robust framework for integrating high-speed HBM3 and HBM4 memory into SoCs using Si & Organic interposer technologies. The innovative methodologies and solutions presented herein pave the way for future developments in high-bandwidth memory interfaces, essential for meeting the growing demands of AI and other data-intensive applications.

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